

Determination of total phenolic content and antioxidant activity of garlic (*Allium sativum*) and elephant garlic (*Allium ampeloprasum*) by attenuated...

The total phenolic contents and antioxidant activities of garlicks from California, Oregon, Washington, and New York were determined by Fourier transform infrared (FT-IR) spectroscopy (400-4000 cm^{-1}). The total phenolic content was quantified [Folin-Ciocalteu assay (FC)] and three antioxidant activity assays, 2,2-diphenyl-picrylhydrazyl (DPPH) assay, Trolox equivalent antioxidant capacity (TEAC) assay, and ferric reducing antioxidant power (FRAP), were employed for reference measurements. Four independent partial least-squares regression (PLSR) models were constructed with spectra from 25 extracts and their corresponding FC, DPPH, TEAC, and FRAP with values for 20 additional extracts predicted ($R > 0.95$). The standard errors of calibration and standard error of cross-validation were <1.45 (TEAC), 0.36 (FRAP), and 0.33 μmol Trolox/g FW (DPPH) and 0.55 mg gallic acid/g FW (FC). Cluster and dendrogram analyses could segregate garlic grown at different locations. Hydroxyl and phenolic functional groups most closely correlated with garlic antioxidant activity.